



Department Of Mechanical Engineering

Complex Engineering Problem CEP

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System Description and Design Parameters:

1. Project Overview

Problem Statement: The objective of this project is to design and simulate an air conditioning system for a residential living space located in **Sialkot, Pakistan**. The project involves calculating both the heating and cooling loads to ensure thermal comfort for the occupants throughout the year, adhering to the requirements of the ME4109: Refrigeration and Air Conditioning CEP.

2. Space Specifications

The selected space is a single zone (Study/Bedroom) located on the ground floor with the following geometric properties derived from the 2D CAD drawings:

- **Room Dimensions:** 17.5 ft (Length) x 12.5 ft (Width) x 10 ft (Height)
- **Total Floor Area:** 219.2 ft²
- **Gross Wall Area:** 600 ft² (calculated from perimeter 60 ft x 10 ft height)

3. Building Envelope & Materials

The room is constructed with specific materials defined to model heat transfer accurately. A key design feature is the distinction between exterior and interior surfaces:

- **Exterior Walls (South & East):**
 - These walls face the outdoors and are subject to direct solar gain and full outdoor temperature differences.
 - **Construction:** Brick with plaster and insulation.
 - **U-Value:** 0.297 BTU/ (hr · ft² · °F).

Wall Assembly Properties - [Outer Wall Thicknesses]

Wall Type

Exterior, Above Grade Wall

Wall Assembly Name:

Outer Wall Thicknesses

Inside Surface Absorptivity:

Light

0.450

Outside Surface Absorptivity:

Medium

0.675

Inside Space	Layer Description	Thickness in	Density lb/cuft	Specific Heat BTU / (lb F)	R-Value (hr sqft F)/BTU	Weight lb/sqft	
▶ Layer #1	1/2-in fiberboard sheathing	0.400	25.0	0.31	0.36900	0.8	
	Layer #2	1/2-in gypsum plaster	0.500	45.0	0.32	0.08500	1.9
	Layer #3	8-in common brick	9.000	120.0	0.20	1.98700	90.0
	Layer #4	1/2-in gypsum plaster	0.500	45.0	0.32	0.08500	1.9
	Outside	Outside surface resistance:				0.16500	
Totals:		10.400			3.37100	94.6	

Overall U-Value: 0.297 BTU/(hr sqft F)

Figure 1- Exterior wall Properties

- **Interior Partition Walls (North & West):**

- These walls face adjacent unconditioned spaces (corridors/rooms), resulting in a lower temperature difference (approx. 50% of exterior ΔT).
- **U-Value:** 0.257 BTU/ (hr · ft² · °F).

Wall Assembly Properties - [Interior Wall Thickness]

Wall Type: Exterior, Above Grade Wall

Wall Assembly Name: Interior Wall Thickness

Inside Surface Absorptivity: Light 0.450 Outside Surface Absorptivity: Medium 0.675

Inside Space	Layer Description	Thickness in	Density lb/cuft	Specific Heat BTU / (lb F)	R-Value (hr sqft F)/BTU	Weight lb/sqft
	Inside surface resistance:				0.68000	
Layer #1	1/2-in fiberboard sheathing	0.400	25.0	0.31	0.36900	0.8
Layer #2	1/2-in gypsum plaster	0.500	45.0	0.32	0.08500	1.9
Layer #3	8-in common brick	9.000	120.0	0.20	1.98700	90.0
Layer #4	1/2-in gypsum plaster	0.500	45.0	0.32	0.08500	1.9
	Outside surface resistance:				0.68000	
	Totals:	10.400			3.88600	94.6

Overall U-Value: 0.257 BTU/(hr sqft F)

Figure 2- Interior wall Properties

- **Roof:**

- Flat roof construction exposed to the sun.
- **U-Value:** 0.638 BTU/ (hr · ft² · °F).

Roof Assembly Properties - [Roof]

Roof Assembly Name: Roof

Inside Surface Absorptivity: Light 0.450 Outside Surface Absorptivity: Light 0.450

Outside	Layer Description	Thickness in	Density lb/cuft	Specific Heat BTU / (lb F)	R-Value (hr sqft F)/BTU	Weight lb/sqft
	Outside surface resistance:				0.13600	
Layer #1	Built-up roofing	0.400	70.0	0.35	0.28400	2.3
Layer #2	6 in. HW concrete	6.000	140.0	0.22	0.49400	70.0
Layer #3	1/2-in gypsum plaster	0.500	45.0	0.32	0.08500	1.9
	Inside surface resistance:				0.56800	
	Totals:	6.900			1.56700	74.2

Overall U-Value: 0.638 BTU/(hr sqft F)

Figure 3- Roof Properties

- **Floor:**
 - Slab-on-grade floor.
 - **U-Value:** 0.374 BTU/ (hr · ft² · °F).

Floor Assembly Properties - [Floor]

Floor Type: Slab Floor On Grade

Floor Assembly Name: Floor

Inside Surface Absorptivity: Medium 0.675

Inside Space	Layer Description	Thickness in	Density lb/cuft	Specific Heat BTU / (lb F)	R-Value (hr sqft F)/BTU	Weight lb/sqft
	Inside surface resistance:				0.92000	
Layer #1	Carpet with rubber pad, R-0.68	0.500	20.0	0.33	1.23000	0.8
Layer #2	1-in stone	1.000	160.0	0.19	0.08000	13.3
Layer #3	6 in. HW concrete	6.000	140.0	0.22	0.44444	70.0
Soil	Totals:	7.500			2.67444	84.2

Overall U-Value: 0.374 BTU/(hr sqft F)

Figure 4- Floor Properties

- **Fenestration (Openings):**
 - **Windows:** 17 ft² area, Single-glazed (U=1.090).

Window Properties - [Window]

Input Method: ☒ Simple ☐ Detailed

Name: Window

Window Performance:

Overall U-Value: 1.090 BTU/(hr sqft F)

Overall SHGC: 0.600

Overall VT: 0.500

Internal Shade:

Type: None

Figure 5- Window Properties

- **Doors:** 52 ft² total area, Opaque wood/metal construction (U=0.470).

Door Properties - [Wooden Door]

Door Details

Name: Wooden Door

Type: Opaque Door

Opaque U-Value: 0.470 BTU/(hr sqft F)

Figure 6- Door Properties

4. Design Conditions (Sialkot, Pakistan)

The system is designed for the specific climatic conditions of Sialkot, ensuring performance during peak summer and winter months.

2025 ASHRAE Handbook - Fundamentals (IP)

Compare



SIALKOT CITY, PAKISTAN (WMO: 416000)

Lat: 32.5N Lon: 74.533E Elev: 820 StdP: 14.27 Time zone: 5.00 (E05) Period: 2003-2023 Grade: A WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
1	39.4	41.6	34.9	30.6	49.2	37.5	34.0	49.3	7.6	58.5	5.9	59.3	0.0	N/A	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
6	22.3	107.8	73.4	104.6	73.9	101.8	73.9	84.7	93.1	83.8	92.3	83.0	91.5	2.3	N/A
Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
82.8	176.6	89.7	81.7	170.0	89.1	81.0	165.8	88.8	49.3	93.2	48.4	92.2	47.5	91.5	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
			Mean		Standard deviation		n=5 years		n=10 years		n=20 years		n=50 years		
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
8.6	6.0	4.7	DB	35.6	111.7	1.3	2.1	34.6	113.2	33.9	114.4	33.1	115.6	32.1	117.2
			WB	34.9	86.2	1.3	1.0	34.0	86.9	33.2	87.5	32.5	88.1	31.6	88.9

Monthly Climatic Design Conditions

		Annual												Jan		Feb		Mar		Apr		May		Jun		Jul		Aug		Sep		Oct		Nov		Dec	
		DBAvg		74.0	52.8	59.5	68.2	78.8	86.8	89.4	86.3	85.4	83.5	76.1	64.9	55.5	DBStd		13.45	3.85	4.60	5.86	5.76	4.91	5.25	4.01	3.24	3.24	4.67	3.89	4.51						
Temperatures, Degree-Days	HDD50		25	16	1	0	0	0	0	0	0	0	0	0	8	HDD65		920	379	162	33	1	0	0	0	0	0	0	0	51	294						
	CDD50		8781	102	266	564	865	1140	1182	1126	1097	1004	810	446	179	CDD65		4202	0	8	132	416	675	732	661	632	554	345	46	1							
	Wind		WSAvg	0.9	0.5	1.0	1.2	1.4	1.7	1.4	1.0	0.8	0.6	0.5	0.3	0.3																					
	Precipitation	PrecAvg		39.3	1.7	1.7	2.0	1.1	0.9	3.0	12.4	11.0	3.9	0.8	0.4	0.6	PrecMax		74.3	5.6	6.5	12.2	5.6	2.5	7.0	27.3	36.1	22.3	3.2	2.0	2.2						
		PrecMin		21.4	0.0	0.0	0.0	0.0	0.0	0.0	0.3	2.4	1.7	0.2	0.0	0.0	0.0	PrecStd		13.1	1.5	1.7	2.6	1.2	0.7	1.8	5.4	8.5	4.1	1.0	0.6	0.6					
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures		0.4%	DB	71.6	79.5	93.4	104.2	111.2	111.7	104.9	97.5	96.6	93.7	83.8	75.5	MCWB	59.4	63.2	71.3	72.3	72.1	73.6	79.5	82.8	80.5	77.0	67.5	61.2									
			DB	67.8	76.1	89.0	100.8	107.3	108.9	100.4	95.8	94.9	91.5	81.2	72.5	MCWB	56.9	62.4	70.2	70.6	72.0	73.9	80.6	82.6	80.5	75.5	67.0	60.4									
	2%	DB	65.2	73.5	84.8	97.4	104.2	105.9	97.4	94.6	93.3	89.7	78.8	69.9	MCWB	55.2	60.8	67.4	69.4	71.7	74.8	81.3	82.4	80.2	74.5	65.5	58.8										
		DB	62.7	70.7	81.2	93.7	101.4	102.9	95.1	93.1	91.7	87.5	76.1	67.0	MCWB	54.4	59.2	65.9	68.5	71.8	75.3	81.3	82.2	79.5	73.2	63.9	57.6										
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.3	67.0	75.0	76.6	79.6	84.9	85.4	86.1	84.3	80.8	71.9	64.5	MCDWB	68.2	75.3	87.8	95.1	98.5	96.5	93.8	93.6	92.8	89.3	80.6	72.0									
			WB	58.5	64.2	71.9	74.0	77.5	82.8	84.6	84.8	82.9	79.0	69.0	61.9	MCDWB	65.3	73.6	84.9	91.9	96.6	95.0	93.4	92.4	91.5	87.3	77.8	70.0									
2%		WB	56.7	62.3	69.2	72.3	75.8	81.4	83.5	84.1	81.8	77.5	67.0	60.0	MCDWB	63.1	71.5	81.9	90.4	96.2	94.4	92.4	91.8	90.3	85.7	76.3	68.0										
		WB	55.2	60.5	67.1	70.8	74.1	80.3	82.8	83.1	80.9	75.8	65.0	58.2	MCDWB	60.7	69.0	78.6	88.8	94.4	93.7	91.6	90.8	89.3	84.3	74.0	65.9										
Mean Daily Temperature Range		5% DB	MDBR	18.1	20.3	21.6	25.0	25.8	22.3	14.4	12.9	15.0	21.2	22.7	21.2	MCDDBR	23.4	23.9	26.2	30.5	30.4	28.1	18.5	15.4	16.6	21.2	24.5	25.3									
			MCWBR	13.5	12.0	10.5	9.5	8.9	7.8	6.7	6.5	6.4	8.1	12.5	15.1	MCDWB	19.6	21.7	25.0	27.4	26.8	19.7	15.5	14.1	14.8	18.3	23.0	23.8									
			MCWBR	12.5	12.2	11.4	10.2	10.3	8.1	7.0	6.9	6.7	8.2	13.5	15.4																						
		Clear Sky Solar Irradiance	taub	0.443	0.500	0.537	0.566	0.652	0.753	0.861	0.735	0.616	0.548	0.493	0.436	taud	2.022	1.890	1.834	1.798	1.651	1.518	1.415	1.598	1.780	1.828	1.903	2.039									
	Ebn at noon		234	233	236	236	217	195	174	195	213	215	214	227	Edn at noon	44	56	63	69	80	91	101	82	66	58	49	42										
All-Sky Solar Radiation	RadAvg		920	1211	1608	2008	2237	2199	1761	1698	1685	1531	1155	955	RadStd	98	132	126	91	76	132	161	117	91	75	84	73										

Figure 7- Design Conditions

Parameter	Cooling (Summer)	Heating (Winter)
Outdoor Design Temp (t_o)	106.5°F	39.4°F
Indoor Design Temp (t_i)	75.0°F	70.0°F
Design Temperature Diff (ΔT)	31.5°F	30.6°F
Relative Humidity	50%	N/A
Month of Peak Load	July (1600 Hours)	January

Air System Properties - [Default System]

General | System Components | **Zone Components** | Sizing Data | Equipment

☒ Zones
☒ Thermostats
☒ Supply Terminals
☐ Zone Heating Units

Thermostat and Zone Data

☒ All zone Tstats set the same Zone All of 1

Zone Name: All Zones

Cooling T-stat Setpoints: occ. 75.0 F unocc. 80.0 F

Heating T-stat Setpoints: occ. 70.0 F unocc. 65.0 F

T-stat Throttling Range: 1.50 F

Diversity Factor: 100 %

Direct Exhaust Fan KW: 0.0 KW

Shared Data

Thermostat Schedule: Default Fan/Thermostat Schedule

Unoccupied Cooling is: ☒ Available ☐ Not available

OK Cancel Help

Figure 8- Thermostat Details

5. Internal Loads & Ventilation

To simulate real-world usage, the following internal heat gains were accounted for:

- **Occupancy:** 5 People (Activity Level: Office/Study work).
- **Lighting:** 162 Watts
- **Equipment:** 164 Watts (Computers/Misc. electronics).
- **Infiltration:** 11 CFM (Based on air tightness of 0.3 ACH).
- **Ventilation Requirement:** 19 CFM of fresh outdoor air to maintain air quality.